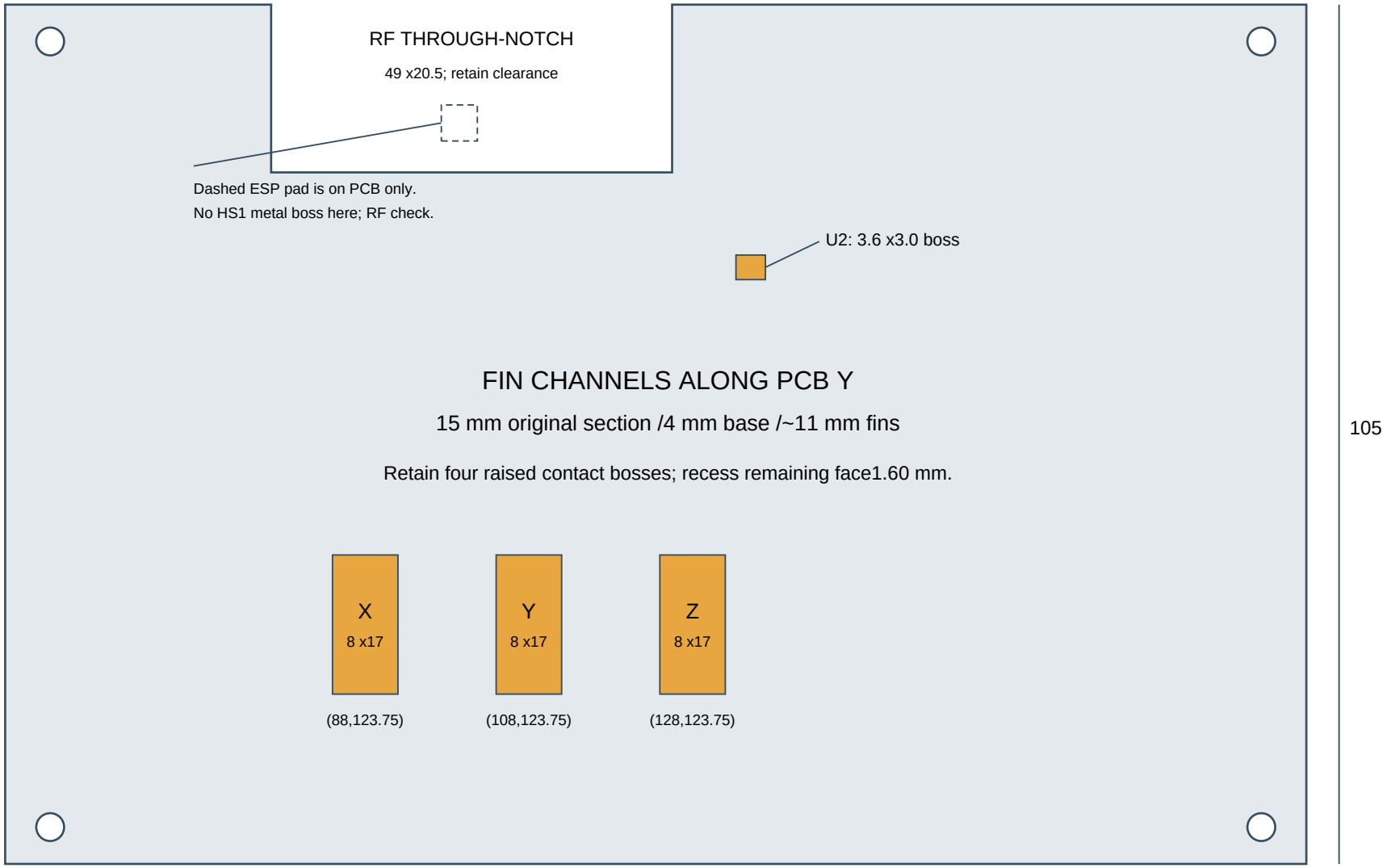
Finned underside cooler - 15.4 mm nominal stack

ABX-HS1-002 | Fischer SK44 150 SA stock, cut to105 mm | Existing four PCB mounting holes

Drawing uses PCB global XY projected through underside (not mirrored). Origin of DXF: X44,Y153.

M3 clearance holes: (49.5,52.5), (197.5,52.5), (49.5,148.5), (197.5,148.5). Diameter3.4.



Machining, TIM and screw assembly

Stock SKU: SK44 150 SA,159 x150 x15 | Trim length to105 | Actual fin pitch follows manufacturer extrusion

Saw/machine stock to159 x105 mm plan; remove the RF notch shown on page1.

Mill1.60 +/-0.03 from the flat face except the three8 x17 driver bosses AND the3.6 x3 U2 boss.

Keep nominal2.40 mm base below the recessed face and approximately11 mm original fins.

Fit four2.00 +/-0.03 hard spacers at H1-H4. No new PCB mounting holes.

Use M3x10 screws, front nylon washers and M3 nuts; mill8 mm fin-side nut/tool recesses.

Check actual fastener stack and mounting-ground bond. Anodizing is not reliable insulation.

Insulating TIM: Bergquist GP5000S35-0.020-02-0404 (TGP5000),0.508 mm sheet.

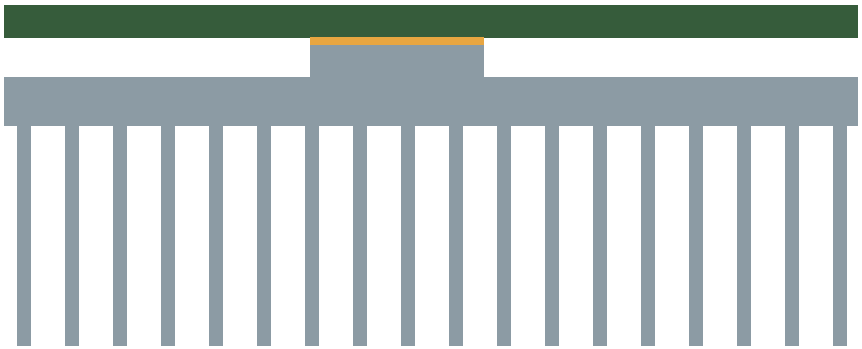
Cut3 x (8 x17) for drivers and1 x (3.6 x3.0) for U2; target0.40 mm assembled thickness.

Contact flatness/coplanarity<=0.05 mm; target Ra<=1.6 um. Check all four contact imprints.

Check board bow<=0.10 mm near contacts. Shim the stack rather than force it with bolt torque.

Trim PTH tails to0.8..1.0 below PCB without cutting solder fillets; inspect every protruding pin.

Section schematic - fin pitch indicative



PCB underside

TIM0.40 / boss rise1.60

Recessed base2.40

Fins ~11.0

Nominal total below PCB15.40

No fan assumed. Prefer vertical fin channels with free inlet/outlet and>=10 mm air beyond fin tips.

The20 mm metal-height allowance does not include enclosure ventilation clearance.

Use the supplied DXF for XY machining; this section does not replace the purchased extrusion drawing.

Exposed pads, RF restriction and thermal findings

CAD improvements preserve all original component positions and routing | Hardware qualification remains pending

ESP: exposed B.Cu GND window4.0 x4.0 at X97.5..101.5/Y60.46..64.46 beneath the module EP.

TIM receiving copper4.4 x4.4. This is a separate-cooler provision, not an HS1 contact boss.

A metal ESP sink here is within the15 mm antenna housing-clearance recommendation.

Keep HS1 antenna notch unchanged; qualify radio range/throughput before fitting a separate ESP sink.

U2: exposed GND window3.2 x2.6 at X133.5..136.7/Y78.8..81.4;3.6 x3.0 receiving copper.

The U2 HS1 boss couples that ground copper through insulating TIM. It does not prove600 mA continuous.

Fabrication: 104 selected thermal holes must be resin filled, planarized and copper capped on both faces.

This includes89 driver vias,12 specifically identified U3 EP41 thermal holes and3 U2 ground stitches.

Do not fill actual connector, header or mounting insertion holes; use the supplied coordinate CSV.

Current thermal screening - all3axes,40 C ambient, hot-max silicon, +5% current

Uses previous8..9 C/W local paths,15% reserve,5 W other heat,10 C allowance and sink target2.5 C/W.

- 2 A RMS command: 2.50 W/driver; 107.0 C screening result.
- 2.5 A RMS command: 3.54 W/driver; 125.6 C screening result.
- 3 A RMS command: 4.79 W/driver; 147.7 C screening result.
- 3.5 A RMS command: 6.23 W/driver; 173.5 C screening result.

2 A remains a conditional prototype target. Higher failed results do not establish physical equilibrium.

Installed2.5 C/W is a design requirement, not a catalogue rating inherited after milling/notching.

Q1 input current and U2 sustained3.3 V load remain important limits; opening copper does not remove them.

Full assumptions and source links: design/thermal-rev1.2-20261023.md

Manufacturer SK44 catalogue: 159 x15 section,4 mm base; use actual supplier extrusion drawing.